



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
UNIVERSITY EXAMINATIONS 2025/2026
THIRD YEAR FIRST SEMESTER EXAMINATION FOR BACHELOR OF SCIENCE
DEGREE IN MECHATRONIC ENGINEERING

EMT 3103 ELECTRICAL MACHINE DRIVES I

DATE: 7/8/2025

TIME: 2-4PM

INSTRUCTIONS

1. This paper consists of FIVE questions.
2. You are required to answer THREE questions only.
3. Question **ONE** is compulsory.
4. Attempt ONE question from **section A** and another ONE question in **section B**.
5. Question ONE carries 30 marks and the others carry 20 marks each.

Question One (30 marks)

- A. State and explain the solution to the problems associated with commutation in a DC machine. [6 marks]
- B. Explain the key difference between a stepper motor and a servo motor based on Feedback mechanism, Torque and speed characteristics and System complexity and cost. [6 marks]
- C. A 4-pole, 900 rpm. DC machine has a terminal voltage of 220 V and an induced voltage of 240 V at rated speed. The armature circuit resistance is 0.2Ω . Is the machine operating as a generator or a motor? Compute the armature current and the number of armature coils if the air-gap flux/pole is 10 mWb and the armature turns per coil are 8. The armature is wave-wound. [5 marks]
- D. State and explain why: [2 marks]
- I. Lap winding is suitable for comparatively low-voltage and high-current generators. [2 marks]
 - II. Wave winding is suitable for high-voltage, low-current machines. [2 marks]
- E. A stepper motor has a step angle of 1.8° and is driven at 4000 pulses per second(pps). Determine: [1 mark]
- I. resolution
 - II. motor speed

- III. number of pulses required to rotate the shaft through 54° . [1 mark]
- F. State and explain the two Transformer tests. [1 mark]
- [6 marks]

SECTION A (Answer only ONE question)

Question Two (20 marks)

- A. Explain the following characteristics of separately excited DC generator:
- No-load characteristic [4 marks]
 - Load/terminal characteristic [4 marks]
- B. With aid of graphs/sketches explain the following characteristics curves of series DC motors:
- Speed versus I_a x-tics [4 marks]
 - Torque versus I_a x-tics [4 marks]
 - Speed versus Torque x-tics (also known as terminal characteristics). [4 marks]

Question Three (20 marks)

- A. A 220 V DC shunt motor has an armature resistance of 0.5Ω and a field resistance of 110Ω . The motor runs at 1000 rpm with an armature current of 20 A. Assuming constant flux, calculate:
- The speed when the armature current is 40A. [5 marks]
 - The gross torque developed at armature currents of 20A and 40 A. [4 marks]
 - The speed when the load torque is halved from its original value at 20A. [5 marks]
- B. A separately excited generator, when running at 1000 rpm supplies 200A at 125V. What will be the load current when the speed drops to 800 rpm if field current remains constant? Given that the armature resistance is 0.04Ω and brush drop is 2V. [6 marks]

SECTION B (Answer only ONE question)

Question Four (20 marks)

- A. A DC motor takes an armature current of 110 A at 480 V. The armature circuit resistance is 0.2Ω . The machine has 6-poles and the armature is lap-connected with 864 conductors. The flux per pole is 0.05 Wb. Calculate the speed and the gross torque developed by the armature. [6 marks]
- B. State and explain the three main types of stepper motors. In your answer, describe the construction and working principle of each type, and discuss their relative advantages and disadvantages in terms of torque, speed, resolution, and detent torque. [9 marks]
- C. A shunt generator delivers 450 A at 230 V and the resistance of the shunt field and armature are 50Ω and 0.03Ω respectively. Calculate the generated e.m.f. [5 marks].

Question Five (20 marks)

- A. Sketch the equivalent circuit of 200/400V, 50HZ, 1-phase transformer from the following test data:
- O.C test: 200V, 0.7A, 70W – on low voltage side.
 - S.C test: 15V, 10A, 85W – on high voltage side

Calculate the secondary voltage when delivering 5kW at 0.8 power factor (p.f) lagging, the primary voltage being 200V.

[8 marks].

- B. With aid of well labelled diagrams, discuss the basic construction of DC Machines.

[12 marks]

$$f_c = \frac{P_{2,110}}{2\pi \Phi_f}$$

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